Power Budget

Team Number:	201
Project Name:	**Fill-A-Bot**
Team Member Names:	**Cole Trask**
Version:	

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or

All Major Components	Component Name	Part Number	SupplyVoltageRange	#	aximumCurre	TotalCurrent(mA)	Unit
	Linear Voltage Regulator	LM7805	5v-18v	1	1000	1000	
	Op Amp	MCP6004	5v	1	60	60	
	PIC Curiosity Nano Board	PIC 18	5v	1	100	100	

+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	aximumCurre	TotalCurrent(mA)	Unit
	Pic Curiosity Nano Board	PIC 18	5v-18v	1	200	200	mA
	Opamp	MCP6004	1v-5v	1	60	60	mA
						0	mA
						0	mA
						0	mA
					Subtotal	260	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	325	mA
c2. Regulator or Source Ch	+5V Regulator	LM7805	(range)	1	1000	1000	mA
			Total Remaining Current Available on +5V Rail			675	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the

External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Outpu	aximumCurre	TotalCurrent(mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	full part numbe	110VAC	+9V	3000	3000	mA
						0	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	(range)	1	1000	1000	mA
						0	mA
			Total Remaining Current Available on External Power Source 1			2000	mA

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit